

# interpolate\_grouped\_time\_series.R

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interpolate\_grouped\_time\_series  
*Interpolate Grouped Time Series*

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## Description

Interpolates grouped numeric time series to a regular age grid.

## Usage

```
interpolate_grouped_time_series(  
  data_time_series = NULL,  
  grouping_variables = "dataset_name",  
  age_variable_name = "age",  
  value_variable_name = "value",  
  interpolation_method = "linear",  
  extrapolation_rule = 1,  
  ties_function = base::mean,  
  age_min = 0,  
  age_max = 12000,  
  time_step = 500,  
  n_cores = 1,  
  show_progress = TRUE  
)
```

## Arguments

`data_time_series`  
A data frame containing grouping, age, and value columns.

`grouping_variables`  
A character vector of column names that define independent time series.

|                                   |                                                                                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>age_variable_name</code>    | Name of the numeric age column.                                                                                                                                    |
| <code>value_variable_name</code>  | Name of the numeric value column.                                                                                                                                  |
| <code>interpolation_method</code> | Interpolation method passed to <code>[stats::approx()]</code> .                                                                                                    |
| <code>extrapolation_rule</code>   | Numeric extrapolation rule passed to <code>[stats::approx()]</code> .                                                                                              |
| <code>ties_function</code>        | Function used by <code>[stats::approx()]</code> to combine tied ages.                                                                                              |
| <code>age_min</code>              | Minimum age in the output grid.                                                                                                                                    |
| <code>age_max</code>              | Maximum age in the output grid.                                                                                                                                    |
| <code>time_step</code>            | Positive spacing between consecutive ages in the output grid.                                                                                                      |
| <code>n_cores</code>              | Number of workers. Use '1' for sequential ' <code>purrr::map()</code> ' execution or a larger integer for parallel ' <code>furrr::future_map()</code> ' execution. |
| <code>show_progress</code>        | Logical indicating whether supported map operations display progress.                                                                                              |

**Details**

Each unique combination of 'grouping\_variables' is interpolated independently with `[stats::approx()]`. Groups that cannot be interpolated return no rows.

Parallel execution temporarily owns the active 'future' plan. The previous plan is restored when the function exits.

**Value**

A data frame containing the grouping columns and interpolated age and value columns.

**See Also**

`[stats::approx()]`

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